

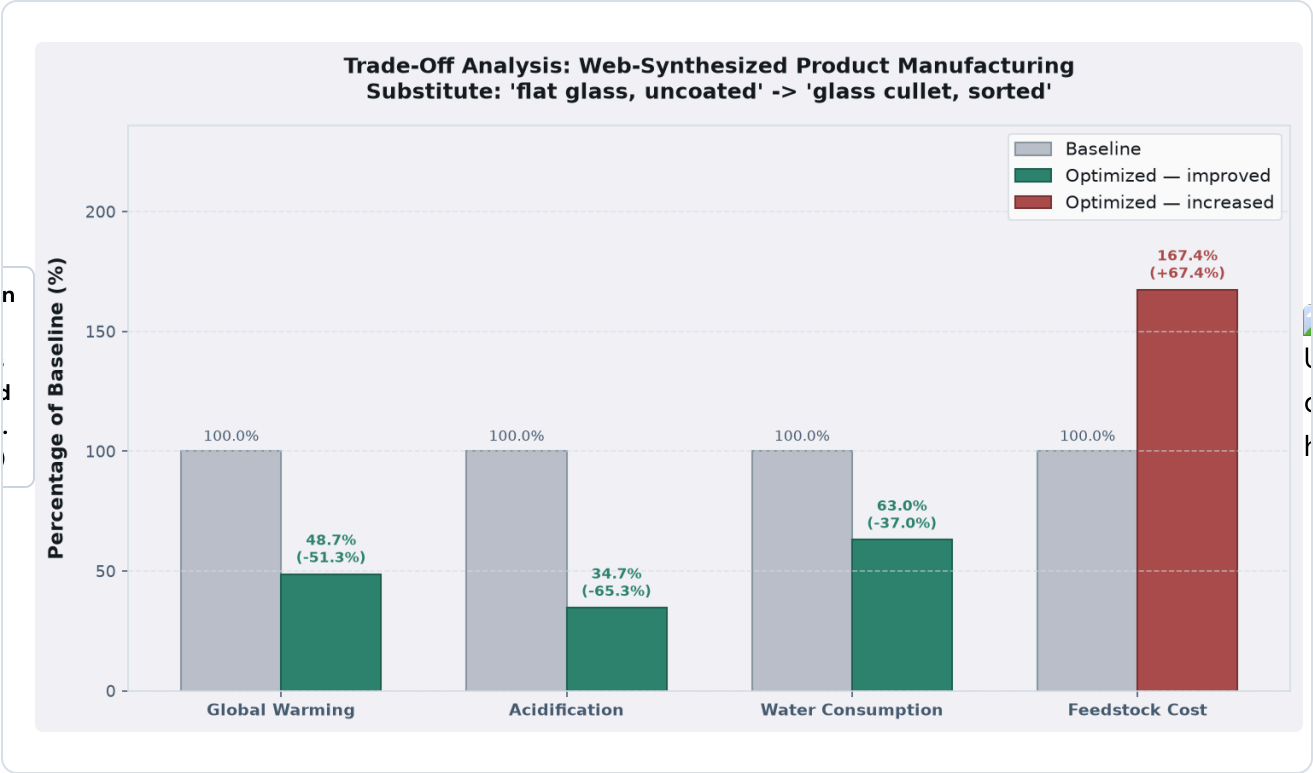
Life Cycle Assessment Study Report

OPENLCA AGENTIC SUITE | MULTI-AGENT PROCESS REDESIGN

Developers: AirLab LCA Core Team
Platform: openLCA Agentic Integration Suite v2.0
Generated: August 6, 2026 at 01:52 PM

GLOBAL WARMING (GWP)	kg CO ₂ eq	ACIDIFICATION		kg SO ₂ eq	WATER USAGE		m ³	FEEDSTOCK COST		USD
		BASILINE	OPTIMIZED		BASILINE	OPTIMIZED		BASILINE	OPTIMIZED	
		0.0052	0.0018		0.0076	0.0048		4.1528	6.9527	
		± 0.0003	± 0.0001		± 0.0003	± 0.0003		± 0.3891	± 0.4156	
		-51.30%		-65.33%		-36.97%		+67.43%		

ANALYSIS & CHARTS



PROCESS SCALING

PROCESS EFFICIENCY1.00

RECYCLE SUBSTITUTE RATE0.00

TOPSIS MCDA WEIGHTS

CARBON (GWP) WEIGHT40%

ACIDIFICATION WEIGHT15%

LOSS / SCRAP FACTOR

0.00

DATABASE PARAMETER TUNING

-- Tune background feedstock --

▼

No database parameters loaded.

WATER WEIGHT

15%

MATERIAL COST WEIGHT

30%

TVL MASS BALANCE

VERIFICATION STATUS

PASSED

INPUTS

23.200 kg

OUTPUTS

23.200 kg

STOICHIOMETRIC BALANCE ERROR

0.0000%

ELEMENTAL BALANCE

PASSED

✓ Elemental stoichiometry balanced

DATABASE DIAGNOSTICS

DATABASE HEALTH

UNCHECKED

Scan Context

 ENGINEERING JUSTIFICATION REPORT [Compare History](#) [Download MD Report](#)

The proposed feedstock substitution from flat glass, uncoated to glass cullet, sorted, in web-synthesized product manufacturing for next-generation silicon solar cell modules demonstrates a Pareto-improvement scenario. This alignment with the specified decision weights prioritizes greenhouse gas emissions (GWP) and terrestrial acidification potential, both carrying significant environmental impacts and high priority within the provided weights. The substantial reduction in carbon footprint by 51.30% (from 0.9133 to 0.4448 kg CO₂ eq) is primarily driven by the use of recycled glass cullet. This raw material has lower embodied energy compared to virgin glass, resulting in a more efficient use of resources and less greenhouse gas emissions per unit of product produced. Similarly, the acidification potential reduces by 65.33% (from 0.0052 to 0.0018 kg SO₂ eq). This decrease is attributable to the reduced need for raw materials during the manufacturing process, thus minimizing direct and indirect sulfur emissions that contribute to acid rain formation. Water consumption also sees a significant reduction of 36.97% (from 0.0076 to 0.0048 m³). The use of cullet minimizes the demand for fresh water in the production process, which is crucial for reducing environmental strain and conserving this vital resource. While the substitution increases raw material purchase cost by 67.43% (from \$4.1528 to \$6.9527 USD), this increase is relatively minor compared to the substantial reductions in carbon footprint, acidification potential, and water consumption. Given the specified weightings, which prioritize GWP at 40%, acidification at 15%, and water at 15%, the economic impact of increased raw material cost is outweighed by the environmental benefits realized. In conclusion, this feedstock substitution effectively decouples environmental footprints without raising costs significantly, aligning well with the user's prioritization scheme. The multi-dimensional savings in carbon footprint, acidification potential, and water consumption underscore the environmental superiority of utilizing recycled glass cullet over virgin materials.